

Q23 Ans: A

To determine the induced e.m.f.:

$\varepsilon = BLv \cos \theta$ where v is the velocity of rod at the bottom of slope, and L is length of rod.

To determine v :

$$v^2 = u^2 + 2as$$

$$v^2 = 0 + 2(g \sin \theta)y$$

$$v = \sqrt{2y(g \sin \theta)}$$

Hence,

$$\varepsilon = BLv \cos \theta = BL \left[\sqrt{2y(g \sin \theta)} \right] \cos \theta = BL\sqrt{2yg} \left[(\sqrt{\sin \theta})(\cos \theta) \right]$$

	$\theta / ^\circ$	Induced emf / V
A	35	0.620 $BL\sqrt{2yg}$ (largest)
B	40	0.614 $BL\sqrt{2yg}$
C	45	0.595 $BL\sqrt{2yg}$
D	50	0.563 $BL\sqrt{2yg}$

Q24 Ans: D